Homogeneous higher order equations

We've already covered solutions of first order and second order differential equations, and in this section of the course we'll be extending what we've learned to higher order equations, like third, fourth, and nth order linear equations.

Solutions to higher order equations

For everything we do with these higher order equations, we'll want to start with them in standard form.

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \ldots + p_1(t)y' + p_0(t)y = g(t)$$

The equation is homogeneous when $g(t) = 0$, and nonhomogeneous when $g(t) \neq 0$.

Homogeneous	$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \ldots + p_1(t)y' + p_0(t)y = 0$
Nonhomogeneous	$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \ldots + p_1(t)y' + p_0(t)y = g(t)$

Notice here that the coefficient on the highest degree derivative $y^{(n)}$ is 1. If we have a higher order equation where that coefficient isn't 1, then we'll want to divide through the equation by that coefficient, to put it in standard form.

Just as with lesser order differential equations, we are guaranteed a solution of the higher order linear differential equation as long as the

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coefficients $p_i(t)$ and the forcing function $g(t)$ are all continuous on the open interval I that contains t_0.

If all the coefficients $p_i(t)$ are continuous on I, if $y_1(t)$, $y_2(t)$, ... $y_n(t)$ are all solutions of the linear differential equation, and if the Wronskian of the solutions $W(y_1, y_2, \ldots y_n)(t) \neq 0$, then the solutions are linearly independent and form a fundamental set of solutions, and the general solution to the homogeneous equation is

$$y(t) = c_1y_1(t) + c_2y_2(t) + \ldots + c_ny_n(t)$$

And the solution to the nonhomogeneous equation will be the sum of the complementary solution $y_c(t)$ (the solution to the associated homogeneous equation) and the particular solution $y_p(t)$.

$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = [c_1y_1(t) + c_2y_2(t) + \ldots + c_ny_n(t)] + y_p(t)$$

Solving higher order homogeneous equations with constant coefficients

Just like before, we'll use a different formula for the general solution of a higher order homogeneous equation depending on the type of roots we find from the associated characteristic equation.

When we have distinct real roots or equal real roots, the solution will be given by these formulas:

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Distinct real roots	$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t} + \ldots + c_k e^{r_k t}$
Equal real roots	$y(t) = c_1 e^{r_1 t} + c_2 t e^{r_1 t} + c_3 t^2 e^{r_1 t} \ldots + c_k t^{k-1} e^{r_1 t}$

But when we have complex conjugate roots from the characteristic equation, the solution depends on the multiplicity of the complex conjugate root pair. As we already know, if the root pair $r = \alpha \pm \beta i$ occurs only once, then $e^{\alpha t}\cos(\beta t)$ and $e^{\alpha t}\sin(\beta t)$ form a fundamental set of solutions, and the general solution is therefore

$$y(t) = e^{\alpha t}(c_1 \cos(\beta t) + c_2 \sin(\beta t))$$

But when the root pair $r = \alpha \pm \beta i$ has multiplicity k, the fundamental set of solutions will be

$e^{\alpha t}\cos(\beta t)$ and $e^{\alpha t}\sin(\beta t)$

$te^{\alpha t}\cos(\beta t)$ and $te^{\alpha t}\sin(\beta t)$

$t^2 e^{\alpha t}\cos(\beta t)$ and $t^2 e^{\alpha t}\sin(\beta t)$

...

$t^{k-1} e^{\alpha t}\cos(\beta t)$ and $t^{k-1} e^{\alpha t}\sin(\beta t)$

Complex conjugate roots always appear in pairs that include both the positive and negative parts of the root pair.

With these forms in mind, let's do an example of a higher order homogeneous differential equation with distinct real roots.

Example

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Find the general solution to the third order homogeneous differential equation.

$$y''' + 2y'' - 13y' + 10y = 0$$

We'll rewrite the differential equation as its associated characteristic equation, and then a calculator to factor the polynomial.

$$r^3 + 2r^2 - 13r + 10 = 0$$

$$(r - 1)(r - 2)(r + 5) = 0$$

From the factored form of the polynomial, we can see that the equation has distinct real roots $r_1 = 1$, $r_2 = 2$, and $r_3 = -5$. And the general solution is therefore

$$y(t) = c_1 e^t + c_2 e^{2t} + c_3 e^{-5t}$$

Keep in mind that we can also solve initial value problems with these higher order differential equations.

We need the same number of initial conditions as the order of the equation. So for a third order equation we'll need three initial conditions, for a fourth order equation we'll need four initial conditions, etc.

Once we have the general solution, we take its derivatives, plug the initial conditions into these, and by doing so we'll generate a system of equations that we can solve for c_1, c_2, c_3, etc. And of course we can solve

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initial value problems this way not only for higher order homogeneous equations with distinct real roots, but also for those with equal real and complex conjugate roots.

Let's do another example, this time with a combination of distinct real roots and equal real roots.

Example

Find the general solution to the fifth order homogeneous differential equation.

$$y^{(5)} + 4y^{(4)} - 6y''' - 32y'' - 35y' - 12y = 0$$

We'll rewrite the differential equation as its associated characteristic equation, and then a calculator to factor the polynomial.

$$r^5 + 4r^4 - 6r^3 - 32r^2 - 35r - 12 = 0$$

$$(r - 3)(r + 4)(r + 1)(r + 1)(r + 1) = 0$$

From the factored form of the polynomial, we can see that the equation has distinct real roots $r_1 = 3$ and $r_2 = -4$, and equal real roots $r_3 = r_4 = r_5 = -1$. The distinct real roots portion of the solution will be $c_1e^{3t} + c_2e^{-4t}$, while the equal real roots portion will be $c_3e^{-t} + c_4te^{-t} + c_5t^2e^{-t}$, and therefore the general solution of the fifth order linear equation is

$$y(t) = c_1e^{3t} + c_2e^{-4t} + c_3e^{-t} + c_4te^{-t} + c_5t^2e^{-t}$$

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Let's do one more example, this time with multiple pairs of complex roots, so that we can see what that looks like.

Example

Find the general solution to the seventh order homogeneous differential equation.

$$y^{(7)} - 2y^{(6)} + 7y^{(5)} - 26y^{(4)} + 29y''' - 60y'' + 100y' = 0$$

We'll rewrite the differential equation as its associated characteristic equation, and then a calculator to factor the polynomial.

$$r^7 - 2r^6 + 7r^5 - 26r^4 + 29r^3 - 60r^2 + 100r = 0$$

$$r(r-2)(r-2)(r^2 + r + 5)^2 = 0$$

From the factored form of the polynomial, we can see that the equation has one distinct real root $r_1 = 0$, two equal real roots $r_2 = r_3 = 2$, and a pair of complex conjugate roots,

$$r_4 = -\frac{1}{2} + \frac{\sqrt{19}}{2}i \text{ and } r_4 = -\frac{1}{2} - \frac{\sqrt{19}}{2}i$$

But because that complex root portion of the factored characteristic polynomial is squared, $(r^2 + r + 5)^2$, these complex roots have multiplicity two, which means we'll have two pairs of complex roots,

$$e^{\alpha t}\cos(\beta t) \text{ and } e^{\alpha t}\sin(\beta t)$$

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$te^{\alpha t}\cos(\beta t)$ and $te^{\alpha t}\sin(\beta t)$

With $\alpha = -1/2$ and $\beta = \sqrt{19}/2$, the complex conjugate solution pairs will be

$$e^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{19}}{2}t\right) \text{ and } e^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{19}}{2}t\right)$$

$$te^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{19}}{2}t\right) \text{ and } te^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{19}}{2}t\right)$$

The distinct real roots portion of the solution will be $c_1e^{0t} = c_1$, the equal real roots portion will be $c_2e^{2t} + c_3te^{2t}$, the complex conjugate roots portion will be,

$$c_4e^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{19}}{2}t\right) + c_5e^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{19}}{2}t\right)$$

$$+c_6te^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{19}}{2}t\right) + c_7te^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{19}}{2}t\right)$$

and therefore the general solution of the seventh order linear equation is

$$y(t) = c_1 + c_2e^{2t} + c_3te^{2t} + c_4e^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{19}}{2}t\right) + c_5e^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{19}}{2}t\right)$$

$$+c_6te^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{19}}{2}t\right) + c_7te^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{19}}{2}t\right)$$

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